

The Best Tools for Backing Up Enterprise Data

To prevent a disastrous loss of data, regular backups are not only recommended but are *de jure*. From the many open source tools available in the market for this purpose, this article helps systems administrators decide which one is best for their systems.



Before discussing the need for backup software, some knowledge of the brief history of storage is recommended. In 1953, IBM recognised the importance and immediate application of what it called the 'random access file'. The company then went on to describe this as having high capacity with rapid random access to files. This led to the invention of what subsequently became the hard disk drive. IBM's San Jose, California laboratory invented the HDD. This disk drive created a new level in the computer data hierarchy, then termed random access storage but today known as secondary storage.

The commercial use of hard disk drives began in 1957, with the shipment of an IBM 305 RAMAC system including IBM Model 350 disk storage, for which a US Patent No. 3,503,060 was issued on March 24, 1970.

The year 2016 marks the 60th anniversary of the venerable hard disk drive (HDD). Nowadays, new computers

are increasingly adopting SSDs (solid-state drives) for main storage, but HDDs still remain the champions of low cost and very high capacity data storage.

The cost per GB of data has come down significantly over the years because of a number of innovations and advanced techniques developed in manufacturing HDDs. The graph in Figure 1 gives a glimpse of this.

The general assumption is that this cost will be reduced further. Now, since storing data is not at all costly compared to what it was in the 1970s and '80s, why should one take backup of data when it is so cheap to buy new storage. What are the advantages of having backup of data?

Today, we are generating a lot of data by using various gadgets like mobiles, tablets, laptops, handheld computers, servers, etc. When we exceed the allowed storage capacity in these devices, we tend to push this data to the cloud or take a backup to avoid any future disastrous events. Many